INCH-POUND
MIL-DTL-32099E(OS)
5 November 2024
SUPERSEDING
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25 March 2011

DETAIL SPECIFICATION

CARTRIDGE, IMPULSE, CCU-145/A

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWC IHD) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the manufacture, assembly, tests upon which product acceptance will be based, and methods of inspection of the CCU-145/A impulse cartridge (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-45444 - Cyclotetramethylenetetranitramine (HMX)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests

MIL-STD-1168 - Ammunition Lot Numbering and Ammunition Data Card

DOD-STD-2101 - Classification of Characteristics

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to usn.ncr.comnavseasyscomdc.mbx.command-standards@us.navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1377

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NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003) DRAWINGS

841AS625	-	Impulse Cartridge, CCU-145/A
841AS627	-	Ignition Charge
841AS628	-	Booster Charge
841AS634	-	Header Assembly
841AS635	-	Eyelet
841AS636	-	Contact Pin
841AS868	-	Vibration Test Fixture
3205AS409	-	CCU-145/A Alternate Test Fixture

(Copies of these documents are available from Commander, Indian Head Division, Naval Surface Warfare Center, CAD/PAD Data Management Branch [Code E26], 4393 Benson Road, BLDG 1557, Indian Head, MD 20640-5092. Refer questions, inquiries, or problems to nswc_iheodtd_dmo@navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQC B1	-	Guide for Quality Control Charts
ANSI/ASQC B2	-	Control Chart Method of Analyzing Data
ANSI/ASQC B3	-	Control Chart Method of Controlling Quality During Production

(Copies of these documents are available online at https://webstore.ansi.org/.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM E1742/E1742M	-	Standard Practice for Radiographic Examination

(Copies of this document are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article tests. When specified (see 6.2), a sample shall be subjected to a first article inspection in accordance with 4.2.

3.2 Conformance to documents. The CCU-145/A impulse cartridge and all its components shall be manufactured in accordance with Drawing 841AS625 and all documents and drawings listed thereon.

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3.3 Materials. Unless otherwise specified (see 6.2), all materials used in the manufacture of the cartridges shall conform to the specifications referred to on the respective Naval Air Systems Command drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture.

3.4 Primary components. For the purposes of this specification, the ignition charge (Drawing 841AS627), the booster charge (Drawing 841AS628), and the main charge consisting of HMX (MIL-DTL-45444, grade B, class 1) are defined as primary components. Only primary components from a single lot shall be used in a production lot of cartridges. One primary component production lot may be used in more than one cartridge production lot.

3.4.1 Ignition and booster composition loading. Composition shall be determined by weight rather than by volume measurement. In addition to supplier certification, new lots of material shall be independently certified to ensure accurate composition and consistent performance.

3.4.2 Main charge (HMX).

3.4.2.1 Quantity. Charges shall be determined by weight rather than by volume measurements.

3.4.2.2 Quality characteristics. The main charge weight shall be a quality characteristic for statistical process control. Sampling for control charting shall be conducted in accordance with established procedures.

3.5 Subassembly requirements.

3.5.1 Insulation resistance. The insulation resistance of the header assembly (Drawing 841AS634) shall be greater than 50 megohms at a relative humidity (RH) of less than 80 percent when measured in accordance with 4.5.1.

3.5.2 Hydrostatic pressure. The header assembly (Drawing 841AS634) shall not leak, crack, or suffer any damage when hydrostatic pressure is applied in accordance with 4.5.2.

3.6 Cartridge requirements.

3.6.1 Visual requirements. All cartridges shall be free of damaged or missing components when examined visually in accordance with 4.5.3. Cartridges that fail to meet this requirement shall be rejected.

3.6.2 Radiographic examination. When radiographically examined in accordance with 4.5.4, any cartridge exhibiting missing, defective, or improperly assembled components, foreign matter, cracks, abnormal voids, or other abnormalities that could adversely affect the form, fit, or function shall be rejected.

3.6.3 Seal integrity. Cartridges that exhibit a leak rate greater than 1.0E-5 standard cc/second of helium when tested in accordance with 4.5.5 shall be rejected.

3.6.4 Bridgewire circuit resistance. The resistance of the bridgewire circuit in each completely assembled cartridge, when measured electrically as specified in 4.5.6, shall be not less than 0.85 ohm and be not greater than 1.15 ohm.

3.6.5 Filter. The output voltage as measured on the oscilloscope shall be less than 77 millivolts peak to peak when tested in accordance with 4.5.7.

3.6.6 Power current. Cartridges shall not initiate or dud when tested in accordance with 4.5.8.

3.6.7 Electrostatic discharge. Cartridges shall not fire, dud, or otherwise be adversely affected when subjected to electrostatic discharge testing in accordance with 4.5.9. After electrostatic discharge testing, cartridges shall meet the bridgewire resistance requirements of 3.6.4 and the ballistic requirements of 3.8.

3.6.8 Stray voltage. Cartridges shall not fire, dud, or otherwise be adversely affected when subjected to the stray voltage test in accordance with 4.5.10. After stray voltage testing, cartridges shall meet the bridgewire resistance requirements of 3.6.4 and the ballistic requirements of 3.8.

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3.7 Environmental conditions.

3.7.1 Forty-foot drop. Cartridges shall not initiate or lose primary components and shall be safe to handle for disposal when tested in accordance with 4.5.11. Cartridges subjected to the 40-foot drop test shall not be ballistically tested.

3.7.2 Six-foot drop. Cartridges shall not fire or incur any detrimental internal or external damage or degradation as a result of the 6-foot drop test in accordance with 4.5.12 and shall meet the requirements of 3.8 when ballistically tested.

3.7.3 Temperature, humidity, and altitude (TH&A) cycling. Cartridges shall not initiate or incur any internal or external damage or degradation when subjected to TH&A cycling in accordance with 4.5.13. After exposure to the TH&A environment, cartridges shall be ballistically fired and shall meet the requirements of 3.8.

3.7.4 Shock. Cartridges shall not initiate or incur any internal or external damage or degradation when subjected to the shock test in accordance with 4.5.14. After exposure to the shock environment, cartridges shall be ballistically fired and shall meet the requirements of 3.8.

3.7.5 Vibration. Cartridges shall not initiate or incur any internal or external damage or degradation when subjected to the vibration test in accordance with 4.5.15. After exposure to the vibration environment, cartridges shall be ballistically fired and shall meet the requirements of 3.8.

3.7.6 Salt fog. Cartridges shall not initiate or incur any degradation when subjected to the salt fog test in accordance with 4.5.16. After exposure to the salt fog environment, the cartridges shall be gently washed in running water that is at standard ambient conditions in accordance with MIL-STD-810, method 509. The cartridges shall be ballistically fired after washing. The cartridges shall meet the requirements of 3.8.

3.8 Ballistic requirements. Cartridges shall meet the requirements of 3.8.1 through 3.8.8 when tested in accordance with 4.5.17.

3.8.1 Projectile velocity. The velocity of the projectile shall be not less than 80 feet per second or greater than 140 feet per second when tested in accordance with 4.5.17.1.

3.8.2 Ignition delay. The time between first application of firing current and first indication of movement of the projectile shall be not more than 50 milliseconds when tested in accordance with 4.5.17.2.

3.8.3 Post-fire resistance. The post-fire resistance shall be greater than 500 ohms in accordance with 4.5.17.3.

3.8.4 Petaling. After functioning, visual examination of the closure disc shall show evidence of petaling.

3.8.5 Crimp integrity. After functioning, the cartridge shall exhibit a secure crimp of 360 degrees between the body and the closure disc. The closure disc shall remain with the cartridge after it is fired.

3.8.6 Misfire. There shall be no misfires.

3.8.7 Mechanical failure. The cartridge shall function in accordance with the performance requirements specified herein without mechanical failure (see 6.6.1). Failure to meet this requirement shall result in rejection of the lot.

3.8.8 Quality characteristics. Ballistic velocity of the projectile shall be a quality characteristic for statistical process control. Sampling for control charting shall be conducted in accordance with established procedures.

3.9 Statistical process control. Statistical process control in accordance with ANSI/ASQC B1, ANSI/ASQC B2, and ANSI/ASQC B3 shall be required for quality characteristics cited within this specification (main charge and ballistic velocity). In addition to other requirements for this specification, only ballistic samples from lots that can demonstrate a state of statistical control shall be submitted for lot acceptance ballistic testing. Samples from lots that cannot demonstrate statistical control shall not be submitted for lot acceptance ballistic tests.

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3.10 Workmanship. Cartridges shall be constructed and finished in a manner to ensure compliance with all the provisions of this specification and the applicable drawings. Special attention shall be directed to dimensions, finishes, coatings, and cleanliness.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Production conformance inspection (see 4.3).

c. Lot acceptance inspection (see 4.4).

4.2 First article inspection. Unless otherwise specified (see 6.2), a first article inspection sample of 87 cartridges shall be prepared. This sample, manufactured and packaged in the same methods, materials and procedures proposed for the production lot, is for the purpose of determining that the contractor, prior to starting production, is capable of producing items that comply with the technical requirements of the contract. A first article sample acceptable for environmental and ballistic testing shall be defined as a sample that has met the requirements of 3.2 through 3.6.8 of this specification. The first article test sample shall be expended in the tests listed in table I. Segregate and serialize the first article test sample in accordance with table II. Failure of any cartridge to comply with the requirements of section 3 shall be cause for rejection of the first article represented.

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TABLE I. First article tests.

Test/Inspection	Requirement paragraph	Test paragraph	Group / quantity of cartridges required							
			A	B	C	D	E	F	G	H
Visual inspection	3.6.1	4.5.3	6	6	18	9	12	12	12	12
Radiographic exam.	3.6.2	4.5.4	6	6	18	9	12	12	12	12
Leak test	3.6.3	4.5.5	6	6	18	9	12	12	12	12
Bridgewire resistance	3.6.4	4.5.6	6	6	18	9	12	12	12	12
Filter test	3.6.5	4.5.7	6	6	18	9	12	12	12	12
Power current	3.6.6	4.5.8	-	-	-	-	-	12	12	12
Electrostatic discharge	3.6.7	4.5.9	6	6	18	9	12	-	-	-
Bridgewire resistance	3.6.4	4.5.6	6	6	18	9	12	12	12	12
Stray voltage	3.6.8	4.5.10	6	6	18	9	12	-	-	-
Bridgewire resistance	3.6.4	4.5.6	6	6	18	9	12	-	-	-
Forty-foot drop	3.7.1	4.5.11	6	-	-	-	-	-	-	-
Six-foot drop	3.7.2	4.5.12	-	6	-	-	-	-	-	-
TH&A cycling	3.7.3	4.5.13	-	-	18	-	-	-	-	-
Shock	3.7.4	4.5.14	-	-	-	9	-	-	-	-
Vibration	3.7.5	4.5.15	-	-	-	9	-	-	-	-
Salt fog	3.7.6	4.5.16	-	-	-	-	12	-	-	-
Radiographic exam.	3.6.2	4.5.4	-	6	18	9	12	-	-	-
Leak	3.6.3	4.5.5	-	6	18	9	12	-	-	-
Bridgewire resistance	3.6.4	4.5.6	-	6	18	9	12	-	-	-
Filter test	3.6.5	4.5.7	-	6	18	9	12	-	-	-
Ballistic test (-65 °F)	3.8	4.5.17	-	2	6	3	4	12	-	-
Ballistic test (70 °F)	3.8	4.5.17	-	2	6	3	4	-	12	-
Ballistic test (200 °F)	3.8	4.5.17	-	2	6	3	4	-	-	12
Discard	-	-	6	-	-	-	-	-	-	-

TABLE II. Serial number assignment.

Group	A	B	C	D	E	F	G	H
Ser. No.	1 – 6	7 – 12	13 – 30	31 – 39	40 – 51	52 – 63	64 – 75	76 – 87

4.3 Production conformance inspection. Production conformance inspection for production lots shall consist of inspection of the characteristics specified on the applicable drawings and the tests and examinations of table III in accordance with the specified requirement. Failure of any cartridge to meet the requirements of table III shall be rejected and removed from the lot.

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TABLE III. Production inspection tests and examinations.

Item	Nature of test	Requirement	Test	Classification (DOD-STD-2101)
Primary component	Single lot verification	3.4	-	(M101)
Header assembly	Insulation resistance	3.5.1	4.5.1	(M102)
Header assembly	Hydrostatic testing	3.5.2	4.5.2	(M103)
Ignition and booster mix	Compounding and loading	3.4.1	-	(M104)
Propellant charge	Type and weight verification	3.4.2.1	-	(M105)
Cartridge	Visual inspection	3.6.1	4.5.3	(M106)
Cartridge	Radiographic inspection	3.6.2	4.5.4	(M107)
Cartridge	Leak	3.6.3	4.5.5	(M108)
Cartridge	Bridgewire circuit resistance	3.6.4	4.5.6	(M109)

4.3.1 Lot size. The lot size shall be as specified (see 6.2). The suggested production lot size is 21,600 cartridges. Unless otherwise specified (see 6.2), the lot shall include five units that will be retained for investigative purposes and the test sample units. Cartridges required for all test purposes are selected from the production lot and will not be applied as a part of the quantity specified for delivery by the contract. Lot numbering shall be in accordance with MIL-STD-1168.

4.4 Lot acceptance inspection. Lot acceptance samples shall be subjected to the inspections and tests specified in table IV. Failure of any sample impulse cartridge to comply with the requirements listed shall be cause for rejection of the lot represented.

TABLE IV. Lot acceptance tests.

Test/Inspection	Requirement paragraph	Test paragraph	A	B	C
Quantity	-	-	⅓ sample	⅓ sample	⅓ sample
Visual inspection	3.6.1	4.5.3	all	all	all
Radiographic exam.	3.6.2	4.5.4	all	all	all
Leak test	3.6.3	4.5.5	all	all	all
Bridgewire resistance	3.6.4	4.5.6	all	all	all
Filter test	3.6.5	4.5.7	all	all	all
Power current	3.6.6	4.5.8	all	all	all
Electrostatic discharge	3.6.7	4.5.9	20	-	-
Bridgewire resistance	3.6.4	4.5.6	all	all	all
Ballistic test (-65 °F)	3.8	4.5.17	all	-	-
Ballistic test (70 °F)	3.8	4.5.17	-	all	-
Ballistic test (200 °F)	3.8	4.5.17	-	-	all

4.4.1 Sample size. The test sample size shall be as specified in table V. The test sample units shall be randomly selected in accordance with ASQ/ANSI Z1.4 at inspection level II. Unless otherwise specified (see 6.2), all samples for test purposes shall be provided at the contractor's expense.

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TABLE V. Production lot test sample size.

Lot size	1-100	101-280	281-500	501-1200	1201-3000	3001-5000	5001-10000
Sample size	12	21	33	54	81	111	180

4.5 Tests.

4.5.1 Insulation resistance. The insulation resistance of the header assembly (Drawing 841AS634) shall be checked on a sample basis with an applied voltage of 500±25 volts direct current (DC) at 70±20 °F and an RH of less than 80 percent. The sample size shall be 1.0 percent minimum of the lot size. Voltage shall be applied between the contact pin (Drawing 841AS636) and the eyelet (Drawing 841AS635). Components that fail to meet the requirements of 3.5.1 shall be rejected.

4.5.2 Hydrostatic test. The header assembly (Drawing 841AS634) shall be tested on a sample basis and shall meet the requirements of 3.5.2 when a 8,000±500-pounds per square inch gauge (psig) hydrostatic pressure is applied to surface A for a minimum of 10 seconds. The sample size shall be 1.0 percent minimum of the lot size. Components that fail to meet the requirements of 3.5.2 shall be rejected. Samples selected for testing shall be considered rejects and discarded after testing.

4.5.3 Visual inspection. All cartridges produced under contract shall be visually examined. All cartridges that fail to meet the requirements of 3.6.1 shall be rejected.

4.5.4 Radiographic inspection. The specified items shall be radiographically examined in accordance with ASTM E1742/E1742M (x-ray). One unit may be used as the penetrometer. Any observable imperfection in components or assembly shall be cause for rejection. All items shall be identified with serial numbers in consecutive numerical order beginning with 0001 in each production lot prior to exposure. The items shall be arranged on boards or trays in consecutive order with any missing serial numbers clearly identified on the radiographic plate. Each radiograph shall carry a permanent identification of the items displayed thereon. The radiographic identification shall include item nomenclature and the Department of Defense Identification Code (DODIC), the contract number, the complete lot number assigned to that lot, and the span of serial numbers displayed. Defective items found during radiographic review are to be marked on the radiograph and, prior to shipment, all defective items shall be removed from the production lot. Radiographs of the entire production lot and a detailed description of the defects of removed items shall accompany the ballistic sample to the activity conducting the production lot acceptance tests. Each cartridge shall be positioned such that the centerline of the cartridge is perpendicular to the radiographic beam. Failure to meet the requirements of 3.6.2 shall result in the rejection of that item.

4.5.5 Leak test. Cartridges shall be placed in the pressure chamber and evacuated to a pressure less than 2 pounds per square inch absolute (psia). Only as many cartridges as can be leak tested within 15 minutes from the time units are removed from the pressure chamber shall be processed. Cartridges shall be slowly pressurized to 30±2 psia of helium and maintain for 60±1 minutes. Following the 1-hour period, the pressure shall be released and the cartridges removed from the pressure chamber. Exterior of test units shall be flushed thoroughly with low-pressure air to remove possible trapped helium. Test cartridges shall be leak tested in a dry gas leak tester. Any cartridge that is not tested within a 15-minute time frame shall be retreated in the helium pressure chamber. If one or more cartridges of the lot acceptance sample leak, then a leak test of each cartridge in the lot shall be required. If a 100-percent leak test of the lot is required, it is at the expense of the contractor. Failure of any cartridge to meet the requirements of 3.6.3 shall result in the rejection of that unit.

4.5.6 Bridgewire circuit resistance. The resistance of the bridgewire circuit shall be measured between the contact pin and the cartridge body by means of an instrument that limits the bridgewire current to a maximum of 25 milliamperes. Failure to meet the requirements of 3.6.4 shall result in the rejection of that item.

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4.5.7 Filter test. A sinusoidal waveform shall be input with a signal generator at 1.0±0.1 megahertz for a maximum of 10 milliseconds through an equivalent 47±4.7-ohm resistor 0.25 watt to a 1.0±0.15-ohm load resistor. The signal generator shall be adjusted to obtain an output of 100±5 millivolts peak to peak as measured across the load resistor using an oscilloscope through a 50-ohm termination. The load resistor shall be removed and the cartridge installed. The waveform shall be reapplied and the voltage across the bridgewire noted. Units shall be conditioned at 70±10 °F prior to conducting the test. The units that do not meet the requirement of 3.6.5 shall be rejected. Figure 1 is a schematic of the test setup where R = 47 ohms.

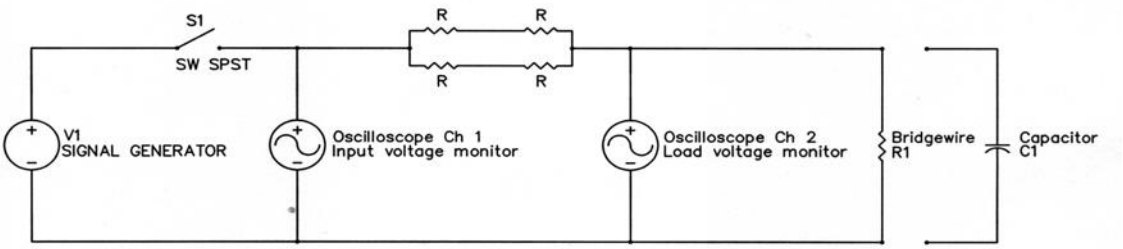


FIGURE 1. Circuit diagram of filter test setup.

4.5.8 Power current. A direct current of not less than 1 ampere supplying a 1-watt minimum shall be applied to the test cartridge bridgewire for a minimum of 5 minutes. The test current shall be regulated throughout the period of application to within 2 percent. If a rectified current is used, the ripple content shall not exceed 5 percent rms of the test current. Units shall be conditioned at 70±10 °F prior to conducting the test. Failure of any cartridge to meet the requirements of 3.6.6 shall be cause for rejection of the lot. The power current test is considered a destructive test and the test shall only be conducted by the designated test activity on the first article and production lot samples. The test shall not be conducted on units intended for delivery to the fleet.

4.5.9 Electrostatic discharge test. Test cartridges shall be connected in a series circuit to a 500±25-picofarad capacitor charged to 25,000±500 volts and a 5,000±250-ohm resistor. The total circuit inductance shall not exceed 5 microhenrys. After assembling the test circuit, a switch shall be closed allowing the capacitor to discharge through the resistor and test cartridge. Units shall be conditioned at 70±10 °F prior to conducting the test. Failure of any cartridge to meet the requirements of 3.6.7 shall be cause for rejection of the lot.

4.5.10 Stray voltage test. Test cartridges shall be subjected to 2,000 pulses of DC voltage with amplitude of 100 ±5 milliamperes. Each pulse shall be 300 milliseconds long and shall be followed by a 200-millisecond quiescent period (i.e., the pulse rate shall be two pulses per second). Units shall be conditioned at 70±10 °F prior to conducting the test. Failure of any cartridge to meet the requirements of 3.6.8 shall be cause for rejection of the lot.

4.5.11 Forty-foot drop test. Test cartridges shall be individually dropped once each from a height of 40 feet onto a 2-inch minimum thick steel plate embedded in concrete. An equal number of cartridges in a sample shall be tested in the following drop orientations: output end up, output end down, and centerline horizontal. A guidance system shall be employed to ensure proper impact in the orientations indicated. Failure of any cartridge to meet the requirements of 3.7.1 shall be cause for rejection of the lot represented.

4.5.12 Six-foot drop test. Test cartridges shall be individually dropped once each from a height of 6 feet +2 inches/ –0 inches onto a 2-inch minimum thick steel plate embedded in concrete. An equal number of cartridges in a sample shall be tested in the following drop orientations: output end up, output end down, and centerline horizontal. A guidance system shall be employed to ensure proper impact in the orientations indicated. Failure of any cartridge to meet the requirements of 3.7.2 shall be cause for rejection of the lot represented.

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4.5.13 TH&A cycling test. The cartridge shall be placed on screen trays or in wire baskets to expose all surfaces and allow air to circulate freely. The cartridges then shall be subjected to the 28-day TH&A cycling test of table VI. This schedule shall be followed for a total of 4 weeks (28 days) except that on the second and fourth weekends, the soak time shall be from 1200 on Friday until 0800 on Monday at a temperature of 160 °F at 95 percent RH. There shall be no interim withdrawals of cartridges. Failure of any cartridge to meet the requirements of 3.7.3 shall be cause for rejection of the lot represented.

TABLE VI. TH&A cycling schedule.

Monday	800	Place test items in a chamber maintained at 70 °F and 50 percent RH.
	1200	Raise chamber temperature to 160 °F and the RH to 95 percent. The chamber temperature shall reach 160 °F at 95 percent RH no later than 1300.
	1600	Remove test items from the chamber and immediately place in a chamber maintained at -65 °F and a pressure altitude of 50,000 feet (1.68 psia).
Tuesday	800	Remove test items from the chamber and immediately place in a chamber maintained at 70 °F and 50 percent RH.
	1200	Remove test items from the chamber and immediately place in a chamber maintained at -65 °F and a pressure altitude of 50,000 feet (1.68 psia).
	1600	Remove test items from the chamber and immediately place in a chamber maintained at 160 °F and 90 percent RH.
Wednesday	800	Reduce chamber temperature to 70 °F and RH to 50 percent. The chamber temperature shall reach 70 °F and 50 percent RH no later than 0900.
	1200	Raise chamber temperature to 160 °F and RH to 95 percent. The chamber temperature shall reach 160 °F and 95 percent RH no later than 1300.
	1600	Remove test items from the chamber and immediately place in a chamber maintained at -65 °F at a pressure altitude of 50,000 feet (1.68 psia).
Thursday	800	Remove test items from above chamber and immediately place in a chamber maintained at 70 °F at 50 percent RH.
	1200	Remove test items from the chamber and immediately place in chamber maintained at -65 °F and a pressure altitude of 50,000 feet (1.68 psia).
	1600	Remove test items from the chamber and immediately place in a chamber maintained at 160 °F and 95 percent RH.
Friday	800	Reduce chamber temperature to 70 °F and RH to 50 percent. The chamber temperature shall reach 70 °F and 50 percent RH no later than 0900.
	1200	Raise chamber temperature to 160 °F and RH to 95 percent. The chamber temperature shall reach 160 °F at 95 percent RH no later than 1300.
	1600	Remove test items from the chamber and immediately place in a chamber maintained at -65 °F and standard ambient pressure.

4.5.14 Shock test. Test cartridges shall be mounted into the test fixture (Drawing 841AS868) cavity and the fixture cap screws torqued to 85 inch-pounds. Test cartridges shall be subjected to a shock test in accordance with method 516, procedure I of MIL-STD-810. A total of 18 terminal peak sawtooth shock pulses, see figure 2, shall be performed with three shocks applied in each direction along the cartridge's three mutually perpendicular axes. The nominal peak amplitude (P) shall be 20g and the nominal duration (D) shall be 11 milliseconds. Failure of any cartridge to meet the requirements of 3.7.4 shall be cause for rejection of the lot represented.

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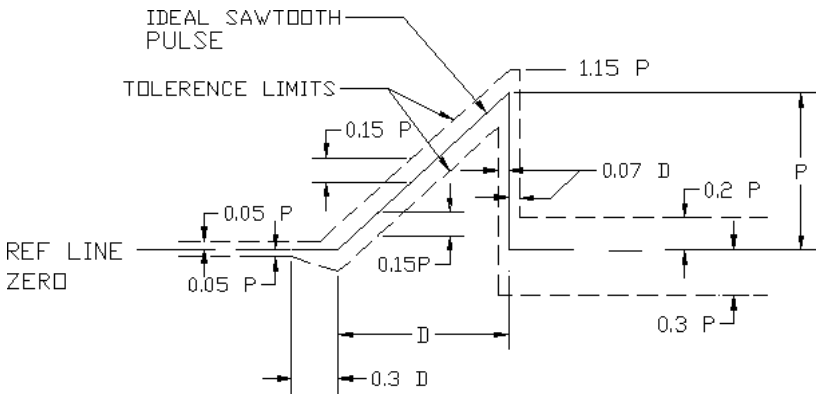


FIGURE 2. Sawtooth shock pulse.

4.5.15 Vibration test. The test cartridges shall be mounted into the vibration test fixture (Drawing 841AS868) cavity and the fixture cap screws torqued to 85 inch-pounds. The input control vibration sensor shall be rigidly attached to the test fixture. The test cartridges shall be subjected to the random performance and endurance vibration test as indicated herein. Failure of any cartridge to meet the requirements of 3.7.5 shall be cause for rejection of the lot.

4.5.15.1 Performance vibration. The cartridges shall be subjected to random performance vibration testing in accordance with the vibration profile of figure 3 along the cartridge's three mutually perpendicular axes. The vibration time per axis is 30 minutes and shall be equally divided across conditioning temperatures of -65 °F, 70 °F, and 200 °F.

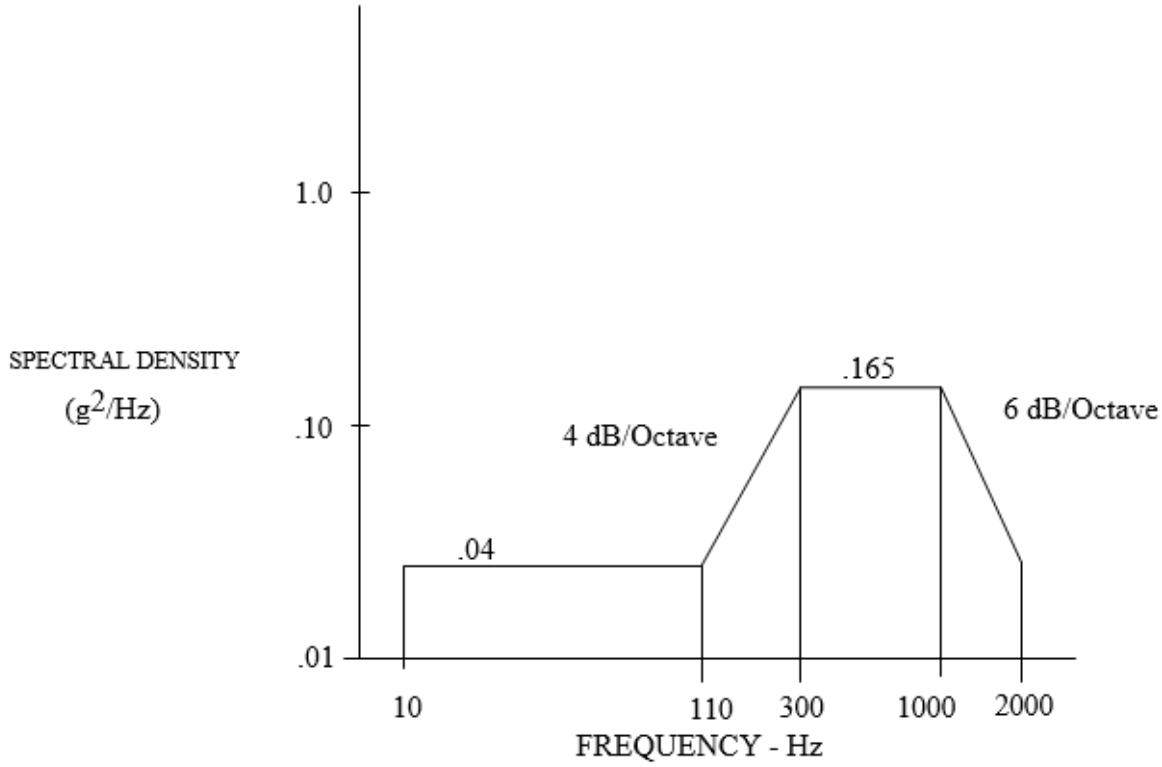


FIGURE 3. Random performance vibration test level.

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4.5.15.2 Endurance vibration. The cartridges shall be subjected to random endurance vibration testing in accordance with the vibration profile of figure 4 along the cartridge's three mutually perpendicular axes. The vibration time per axis is 120 minutes and shall be equally divided across conditioning temperatures of -65 °F, 70 °F, and 200 °F.

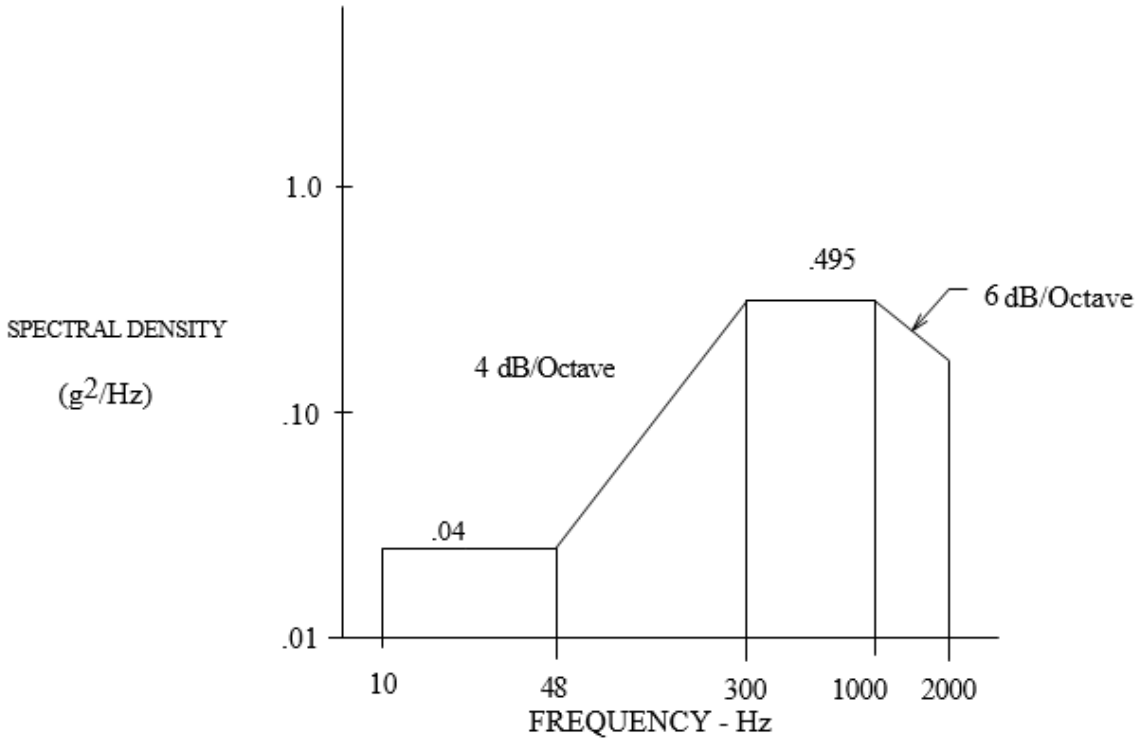


FIGURE 4. Random endurance vibration test levels.

4.5.16 Salt fog test. Test cartridges shall be subjected to a salt fog test in accordance with MIL-STD-810, method 509 with a 5-percent salt concentration for 48 hours. Failure of any cartridge to meet the requirements of 3.7.6 shall be cause for rejection of the lot.

4.5.17 Ballistic tests. Sample cartridges shall be randomly selected to be representative of the complete lot. One third of samples shall be temperature conditioned at –65±5 °F, one-third at 70±5 °F, and one third at 200±5 °F for a minimum of 4 hours but not to exceed 24 hours. The 24-hour maximum does not apply to the 70 °F conditioning. Each sample shall be fired in the test fixture described in 3205AS409 within 3 minutes after removal from conditioning. If the 3-minute limit is exceeded, the unit shall be returned to a 70±5 °F environment for a minimum of 4 hours and then reconditioned in the intended environment for the designated time. A 4.25±0.1-ampere firing current shall be applied for a maximum of 15 milliseconds to the cartridge bridgewire circuit. Failure of any cartridge to meet the requirements of 3.8 shall be cause for rejection of the lot.

4.5.17.1 Projectile velocity. Instrumentation shall be provided to measure the velocity of the projectile ejected from the test fixture. Projectile velocity shall be measured between two induction coils separated by 12.000±0.005 inches located approximately 2 to 3 feet of distance away from the end of the breech. The projectile shall weigh 0.628±0.005 pound and have an outer diameter of 1.000±0.001 inches. The test fixture breech shall have an initial free volume of 0.897±0.050 cubic inch when the projectile is fully inserted. Failure of any cartridge to meet the requirements of 3.8.1 shall result in rejection of the lot.

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4.5.17.2 Ignition delay. Instrumentation shall be provided to measure the time between first application of firing current and the initial movement of the projectile from the breech muzzle. Failure of any cartridge to meet the requirements of 3.8.2 shall result in rejection of the lot.

4.5.17.3 Post-fire resistance. After ballistic firing, the resistance between the contact pin and the cartridge body shall be measured using a certified circuit igniter tester. Failure of any cartridge to meet the requirements of 3.8.3 shall be cause for rejection of the lot.

4.5.17.4 Test failure. Any failures or defects found during lot acceptance testing may be cause for rejection of the lot represented. If test failure is not attributed to the cartridge, the data may be disregarded for that firing and a retest permitted with a new cartridge.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The CCU-145/A impulse cartridge is designed to provide the power to eject expendable countermeasures from aircraft. The impulse cartridge is capable of being a one-for-one replacement of the BBU-35/B impulse cartridge (Air Force Drawing 7729436). The impulse cartridge is designed for use on military aircraft only and, thus, has no commercial application. Current stock of CCU-145/A impulse cartridges may be used until depleted; future requisitions for CCU-145/A impulse cartridges should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Whether first article inspection is required and, if so, specify the test activity (see 3.1 and 4.2).

c. When alternate materials or methods of manufacture are specified on the drawings, the bidders' selections should be clearly stated in the proposal (see 3.3)

d. Production inspection lot size, investigative sample, and test activity (see 4.3.1).

e. Responsibility for providing samples for lot acceptance inspection testing (see 4.4.1).

f. Packaging requirements (see 5.1).

g. Information to be provided by contractors or contracting officers (see 6.4).

h. Applicable safety precaution requirements (see 6.5.4).

6.3 First article inspection. A first article inspection may be deemed necessary when one or more of the following applies: (a) 3 years or more since the impulse cartridge or similar item was last produced, (b) there has been a significant turnover in personnel, (c) there have been major changes to the production facility, (d) there has been a transfer of production to a new facility, or (e) there has been significant change in the design of the impulse cartridge. The cartridge should be delivered to the activity designated by the contracting agency for test. Any production prior to notification by the contracting activity of the first article sample acceptability is at the contractor's risk.

6.4 Data. For the information of contractors and contracting officers, any data specified in (a) subparagraphs below, (b) applicable documents listed in section 2 of this specification, or (c) referenced lower-tier documents need not be prepared for the Government unless specified (see 6.2).

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6.4.1 Standard deliverables. In addition to the delivery of cartridges, the contract may require delivery of the following:

a. Radiographic plates.

b. Radiographic review report.

c. Radiographic sample plates and techniques.

d. Requests for waiver/deviation and engineering change proposal.

e. Progress and management report.

f. Test and inspection report if contractor tested.

g. Ammunition data cards.

h. Performance oriented packaging test report.

i. Process control charts.

6.5 Additional data.

6.5.1 Closed bomb performance. The CCU-145/A impulse cartridge should provide 450- to 750-psig peak pressure at –65 to 200 °F in a 43.5±1.5-cubic centimeter closed vessel. The slope of the output trace should be a maximum of 150 psi per millisecond when measured between 10 to 90 percent of the peak pressure.

6.5.2 All-fire performance. The CCU-145/A impulse cartridge has demonstrated a minimum all-fire level of 2.9957 amperes (Franklin Bruceton Test Method) with a 15-millisecond current pulse duration.

6.5.3 No-fire performance. The CCU-145/A impulse cartridge has a demonstrated a maximum no-fire level of 1.1511 amperes (Franklin Bruceton Test Method) with a 5-minute current pulse duration.

6.5.4 Safety precautions. That the safety precaution requirements of DoDM 4145.26, DoD Contractor's Safety Manual for Ammunition and Explosives, are applicable. NOTE: When this specification is used as a part of the description of work to be accomplished by a Government activity, the safety precautions requirements of NAVSEA OP 5, Ammunition and Explosives Safety Ashore, are applicable (see 6.2).

6.6 Definitions.

6.6.1 Mechanical failure. Any deformation or breakage of a part or separation of parts that is other than a design function is defined as a mechanical failure.

6.7 Subject term (keyword) listing.

Countermeasures

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – OS
(Project 1377-2023-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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